

veys. To ensure compatibility with standard occupational classifications, we translated these estimations to the SOC-18 classification using O\*NET’s crosswalks for a total of 759 occupations.

To enhance our understanding of the GENOE’s effectiveness and to validate our methodology, we conducted several additional exercises. Firstly, we measured task-level exposure by assessing the proportion of tasks at risk within each occupation, allowing us to identify roles where AI might significantly impact certain tasks without necessarily replacing the entire job. We also evaluated tasks in isolation and compared the results with our holistic occupation-level approach to understand the value of considering tasks within their broader occupational context. Additionally, we tested the effect of withholding the occupation name from the model to gauge the importance of occupational context in the AI’s assessment. To incorporate broader societal considerations, we explicitly prompted the model to consider ethical and social factors in its assessments.

Furthermore, we conducted follow-up surveys, asking the AI to explain its reasoning behind the exposure estimates, effectively “opening the black box” of its decision-making process. These additional analyses serve multiple purposes: they validate our methodology by providing alternative perspectives on AI exposure, offer insights into the factors influencing AI’s potential impact on various occupations, and help us understand the decision-making process of the AI model. This approach not only strengthens the robustness of our GENOE index but also potentially reveals any biases or limitations in the AI’s assessments, providing a more reliable picture of occupational exposure to AI.

To further validate our approach, we applied our synthetic survey methodology to replicate the Felten *et al.* (2021) index of occupational exposure. In this exercise, we substituted the original human expert surveys with synthetic surveys conducted using ChatGPT4o. The primary objective was to assess the capability of our AI-driven methodology to reproduce results comparable to those obtained through traditional human expert assessments. This replication serves as a crucial test of our approach’s reliability and accuracy, offering insights into the potential of large language models to emulate human expertise in evaluating occupational exposure to AI.

### 3 RESULTS

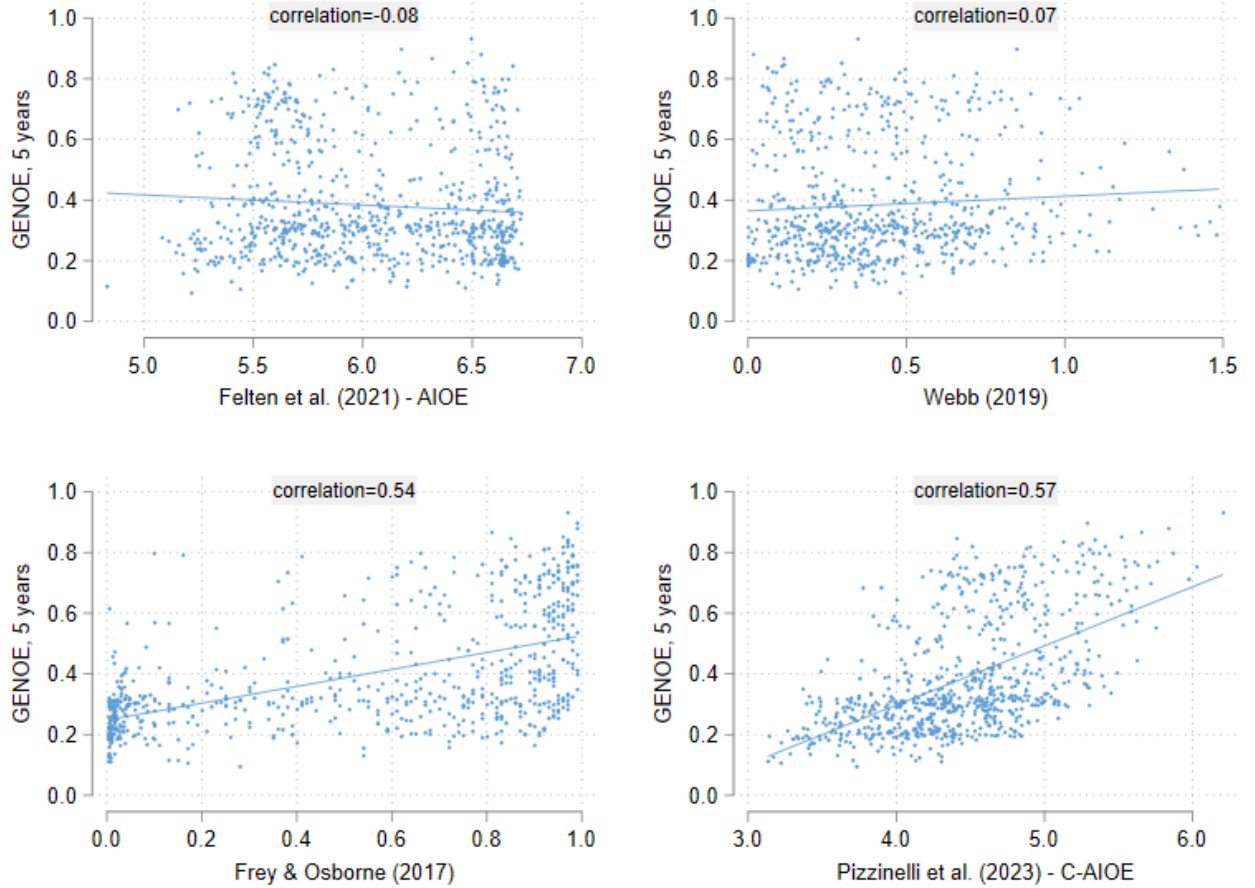
#### 3.1 Baseline Estimations

Figure 1 presents a comprehensive view of AI exposure across 759 occupations, illustrating the average, median, 25th percentile, and 75th percentile for the three defined time horizons. The analysis reveals a progressive increase in average AI exposure: 0.28 within one year, rising to 0.38 over five years, and reaching 0.44 over a decade. This pattern suggests that as AI technologies advance and mature, their potential to reshape various professions grows significantly.<sup>5</sup>

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<sup>5</sup>Complete descriptive statistics can be found in Appendix Table A.1.

**Figure 2: Correlation of GENOE and Other Indexes of Occupational Exposure**



Source: Authors' formulation based on data from O\*NET Felten *et al.* (2021); Webb (2019); Pizzinelli *et al.* (2023). Note: This figure displays the GENOE for the next one, five, and 10 years. The index measures the likelihood that occupations will be displaced by AI within each time horizon.

### 3.3 Alternative Methodologies

We explore how GENOE estimates vary under different methodologies and prompt choices. Our baseline specification involves providing the model with the occupation name and a vector of tasks and asking for an estimate of the likelihood of job replacement. We then introduce six alternative indexes, each employing a distinct approach:

*Blind Index:* This approach removes the occupation name, relying solely on the task vector for estimations. This allows us to isolate the impact of task-specific factors from occupation-specific context factors in the model's predictions and analyze if the absence of the occupation name significantly alters the estimates.

*Task-by-Task Index:* Instead of aggregating tasks under an occupation, we estimate the likeli-